TEXORA · LECTURE SHEET

Fabric Dimensions Measurement

Length, width, thickness and thread density are the first numbers recorded for any fabric, because every later test is cut, counted or calculated from them. This sheet covers measurement on the inspection table, thickness under a defined pressure, and counting of ends, picks, courses and wales.

TE-3201 · QUALITY EVALUATION OF FABRIC AND APPAREL · TOPIC 1 · LEVEL 3 TERM 2

1. Why dimensions come before every other test

A fabric is not tested as an abstract material. A tensile strip is cut to a stated width, a bursting specimen covers a stated area, a stiffness specimen has a stated length, and a mass-per-unit-area result is meaningless unless the specimen was measured accurately. If the width is wrong, the number of ends in a strip is wrong. If the thickness is wrong, bending and compression behaviour will be misread. If the thread density is wrong, cover, mass and strength predictions all shift together. Dimensional measurement is therefore not a preliminary formality; it is the foundation on which the rest of fabric evaluation stands.

Three further reasons make dimensions commercially decisive. First, the buyer buys fabric by length and usable width: a roll that is short or narrow is short delivery, even if its strength is excellent. Second, garment cutting is planned for a stated width; a fabric that varies across or along its length wastes marker efficiency and can make panels impossible to cut. Third, dimensions change with processing — weaving, dyeing, finishing, washing and pressing all alter length, width and thickness — so dimensional values identify where a fabric sits in its production history.

All measurements in this sheet are made on fabric conditioned in the standard atmosphere for textile testing, 65% relative humidity and 20 °C, laid flat and allowed to relax without tension. A fabric measured straight off a roll, still stretched from winding, or in a hot and humid room, will not give the value it will have in use.

2. Length and width on the inspection table

2.1 Principle and apparatus

Length and width are measured on a flat inspection table whose top is smooth, marked with a calibrated scale, and long enough for the fabric to lie without folds or overhang at the measuring zone. The fabric

is drawn over the table without stretching, allowed to rest, and then measured with a rigid rule or a calibrated steel tape. For roll length, counting and measuring machines may be used in production, but the laboratory check remains a direct table measurement over stated portions.

2.2 Procedure for width

- Lay the fabric flat, selvedges parallel, with no applied tension and no creases across the measuring line.
- Measure from selvedge to selvedge, perpendicular to the selvedges. State clearly whether the result is total width or usable width excluding selvedges, because specifications differ.
- Take at least five readings distributed along the length, avoiding the first and last metre of a piece, where tension history is abnormal.
- Record each reading, then report the mean, the minimum and the range. The mean alone hides a narrow place that may fail in cutting.

WORKED EXAMPLE — FABRIC WIDTH

Five width readings on a conditioned plain-weave fabric were 148.6, 149.2, 148.8, 149.0 and 148.9 cm.
Sum = 148.6 + 149.2 + 148.8 + 149.0 + 148.9 = 744.5 cm.
Mean width = 744.5 / 5 = **148.9 cm**.
Range = 149.2 − 148.6 = 0.6 cm. Against a nominal width of 149 cm the fabric is essentially on width, and the small range shows width control was even along the piece. A specification written as 149 ± 1 cm would be met by every reading, not only by the mean.

2.3 Procedure for length

For a cut length, mark two lines square across the fabric a known distance apart under relaxed conditions, or lay the full piece along the table scale without tension and read its end positions. For long pieces, measure in successive table lengths, marking the end of each section before moving the fabric, and add the sections. Never pull the fabric to the next mark: the error introduced by even modest tension in a light fabric easily exceeds the tolerance on length. Report length to the nearest 0.1 cm for laboratory lengths, and state any correction applied for a measuring machine used in bulk checking.

Bowing and skew are checked at the same time, although they are shape faults rather than dimensions in the strict sense. A weft thread that curves across the width (bow) or runs at an angle (skew) changes the effective usable length of a cut panel, so the inspector notes them while the fabric is on the table.

3. Fabric thickness — principle and ISO 5084

Thickness is the distance between the two faces of a fabric when it is held under a defined pressure between a reference plate and a parallel presser foot. The qualification matters: a textile is compressible, so it has no single thickness independent of pressure. A lofty knitted fabric may read 3 mm under a light foot and less than 2 mm under a heavy one. ISO 5084 therefore fixes the conditions — the presser-foot area, the applied pressure, and the time under pressure — so that results from different laboratories can be compared. The usual laboratory arrangement uses a presser foot of defined area (commonly 2000 mm²) and a pressure chosen for the fabric type (commonly 1 kPa for most apparel fabrics), with the reading taken after the specified dwell time.

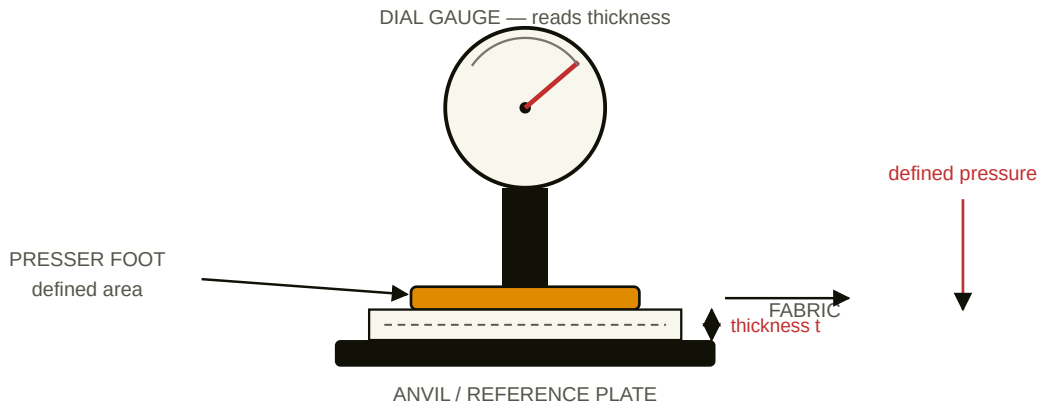


Figure 1 — Thickness gauge in cross-section. The fabric rests on the anvil; the presser foot of defined area loads it at the defined pressure and the dial reads the foot-to-anvil distance.

3.1 Procedure

- Condition the fabric, then place it flat on the anvil without stretching or creasing, in a region away from selvedges and obvious faults.
- Lower the presser foot gently so that it does not impact the fabric, apply the specified pressure, wait the specified time, and read the gauge to 0.01 mm.
- Take at least five, preferably ten, readings at different positions and report the mean. Pile, looped and embossed fabrics need particular care in choosing representative positions.
- Report foot area, pressure and dwell time with the result. A thickness value without its pressure is incomplete.

WORKED EXAMPLE — MEAN THICKNESS

Five readings under the ISO 5084 conditions were 0.42, 0.44, 0.43, 0.45 and 0.43 mm.

Sum = 0.42 + 0.44 + 0.43 + 0.45 + 0.43 = 2.17 mm.

Mean thickness = 2.17 / 5 = **0.434 mm**.

> The spread of only 0.03 mm indicates an even fabric. The same fabric measured under a much higher pressure would give a lower mean, so this value is valid only for the stated pressure and foot area.

Thickness links directly to later topics: bending rigidity rises roughly with the cube of thickness, thermal insulation and air permeability depend on the air held in the thickness, and compression tests in handle evaluation start from the thickness under a low pressure and follow it as pressure rises.

4. Thread counting in woven fabrics

Thread density states how many warp ends and weft picks occupy a unit length, conventionally ends per inch (EPI) and picks per inch (PPI), or ends and picks per centimetre in metric specifications. Density controls cover, mass, strength, stiffness and cost, and it is one of the first checks against a construction specification.

4.1 Counting glass method

A counting glass (linen tester) is a small magnifier mounted over an aperture of known size, commonly 1 inch square or a stated fraction of it. The glass is placed on the relaxed fabric, the threads visible in the aperture are counted — counting a thread that lies half in the aperture by a consistent rule — and the count is repeated at several positions away from the selvedge. For coarse fabrics the aperture count is direct; for fine fabrics a larger measured length is counted by traversing the glass along a straight thread line, or a graticule is used under a low-power microscope.

4.2 Counting by unravelling

Where threads are hard to distinguish — dense, dark, hairy or heavily finished fabrics — a measured width of fabric is marked, the threads are unravelled and counted directly, and the count is divided by the measured width. The method is slower but unambiguous, and it has a second use: the unravelled threads are the specimens for crimp measurement in Topic 2. When density varies between reed marks or across the width, count at left, centre and right and report the pattern, not only a grand mean.

> **WORKED EXAMPLE — ENDS AND PICKS PER INCH**
>
> Counting-glass readings at five positions gave a mean of 69.6 warp ends in a 1-inch aperture and 58.8 weft picks in a 1-inch aperture.
>
> Reported density: **EPI = 70** (mean 69.6) and **PPI = 59** (mean 58.8), rounded to the nearest whole thread for specification purposes, with the unrounded means kept in the laboratory record.
>
> From these values a 50 mm tensile strip in the warp direction is expected to contain about 138 ends — a calculation used fully in Topic 4.

Care points: count on conditioned, relaxed fabric; do not count in the selvedge; align the aperture edges with the thread directions so diagonal threads are not double counted; and in fabrics with a repeating pattern (for example two thick ends alternating with thin ends) count over a whole number of repeats and divide.

5. Courses and wales in knitted fabrics

A knitted fabric has no ends and picks. Its two densities are courses per unit length (CPC, courses per centimetre, or courses per inch) counted along a wale, and wales per unit length (WPC) counted along a course. The counting glass is used in the same way, with the loops identified row by row; in fine or dark knits a needle is used to follow a course across the counting field. Because knitted fabrics extend easily, the fabric must be fully relaxed — ideally after a stated relaxation treatment — and must not be held stretched on the table, or both counts will read low.

$$\text{Stitch density} = \text{courses per unit length} \times \text{wales per unit length}$$

WORKED EXAMPLE — STITCH DENSITY OF A SINGLE JERSEY

A relaxed single-jersey fabric gave CPC = 46 courses per inch and WPC = 38 wales per inch.
Stitch density = 46 × 38 = **1,748 loops per square inch**.
Combined with loop length, stitch density explains the mass, tightness and dimensional stability of the knit: two fabrics with the same counts but different loop lengths are not the same fabric, which is why knitted specifications state loop length alongside CPC and WPC.

Course and wale counts also expose production faults: barré, uneven feeding and needle damage all appear as local changes in density before they appear as visible stripes, so counting is a diagnostic tool as well as a specification check.

6. Reporting and common errors

QUANTITY	UNIT	REPORT	TYPICAL ERROR
Length	m or cm	Value, method, tension condition	Measuring under winding tension
Width	cm	Mean, minimum, range; total or usable	Single reading at one end of the piece
Thickness	mm	Mean, foot area, pressure, dwell time	Reporting thickness without pressure

Woven density	EPI, PPI (or per cm)	Mean at several positions	Counting in the selvedge
Knitted density	CPC, WPC, loops/unit area	Mean on relaxed fabric	Counting a stretched knit

The general rule for this topic, and for the whole course, is simple: a dimensional result is a mean of repeated readings, taken on conditioned and relaxed fabric, reported with its conditions and its variation. A single unchecked reading is an anecdote, not a measurement.

Glossary

Ends per inch (EPI) — number of warp threads in one inch of fabric width. **Picks per inch (PPI)** — number of weft threads in one inch of fabric length. **Counting glass** — magnifier over a fixed aperture used to count threads. **Presser foot** — the flat plate of defined area that loads a fabric in a thickness gauge. **Course** — a row of knitted loops across the fabric. **Wale** — a column of knitted loops along the fabric. **Stitch density** — loops per unit area, the product of course and wale densities. **Usable width** — width excluding selvedges, available for cutting.

Key points

• Dimensions are measured first because specimen size, thread numbers and area calculations for all later tests depend on them.
• Width is the mean of at least five readings along the piece; report the minimum and range as well.
• Thickness exists only at a stated pressure and foot area — ISO 5084 fixes those conditions.
• Count threads with a counting glass, or by unravelling when threads cannot be distinguished visually.
• For knits, report courses and wales per unit length and their product, stitch density, on fully relaxed fabric.
• All dimensional work is done at 65% RH and 20 °C on fabric laid flat without tension.

Practice

1. Width readings are 152.3, 151.8, 152.0, 152.5 and 151.9 cm. Calculate the mean width and the range, and state whether every reading meets a specification of 152 ± 0.5 cm.
2. Thickness readings are 0.61, 0.58, 0.60, 0.59 and 0.62 mm. Calculate the mean thickness and explain why the pressure and presser-foot area must be reported with it.
3. A knitted fabric has CPC = 52 per inch and WPC = 41 per inch. Calculate its stitch density in loops per square inch and state one property this value helps to explain.

4. A counting-glass aperture is 0.5 inch wide and contains 36 ends. Calculate EPI, and explain when counting by unravelling would be preferred to the counting glass.

Answer key

1. Sum = 152.3 + 151.8 + 152.0 + 152.5 + 151.9 = 760.5 cm. Mean = 760.5 / 5 = **152.1 cm**. Range = 152.5 − 151.8 = **0.7 cm**. The specification allows 151.5 to 152.5 cm; all five readings lie inside it, so every reading meets the specification.
2. Sum = 0.61 + 0.58 + 0.60 + 0.59 + 0.62 = 3.00 mm. Mean = 3.00 / 5 = **0.60 mm**. A fabric is compressible, so its measured thickness falls as pressure rises; without the pressure and foot area another laboratory cannot reproduce the value and the number cannot be compared with a specification.
3. Stitch density = 52 × 41 = **2,132 loops/in²**. It helps to explain, for example, mass per unit area or tightness and cover when combined with loop length.
4. EPI = 36 / 0.5 = **72 ends per inch**. Unravelling is preferred when threads cannot be distinguished reliably through the glass — in very dense, dark, hairy or heavily finished fabrics — because each thread is separated and counted directly.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib